

Coding & Solution

Date	15 March 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	[e.g., Python, JavaScript, Java]
Framework(s) Used	[e.g., Flask, React, Spring Boot]

Key Features Implemented	• HDI dataset preprocessing • Data visualization (heatmap, pair plots, histograms) • Feature selection and scaling • Logistic Regression, Decision Tree, and Random Forest models • Model evaluation using Accuracy, Confusion Matrix, and Classification Report • HDI category prediction (Low, Medium, High, Very High) • Flask-based web application with user-friendly interface • Model saved using Pickle for deployment
Pending / Incomplete Features	• User authentication (optional) • Online deployment on Render/Heroku (optional) • Dashboard with interactive charts (future enhancement)
Setup / Run Instructions	1. Clone the repository. 2. Install required packages using pip install -r requirements.txt. 3. Train the model using python train.py. 4. Start the Flask application using python app.py. 5. Open http://127.0.0.1:5000 in a web browser. 6. Enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per capita to predict the HDI category.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes

5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (<i>after uploading to GitHub</i>)

Additional Notes / Comments:

- The project follows a modular architecture with separate files for preprocessing, model training, prediction, and the Flask application.
- Machine learning models are evaluated, and the best-performing model is saved using Pickle.
- The web interface enables users to input HDI indicators and receive predictions instantly.
- The project adheres to clean coding practices, including proper naming conventions, comments, reusable modules, and structured folder organization.
- The solution can assist researchers, policymakers, and educational institutions in analyzing and predicting Human Development Index categories based on socio-economic indicators.